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Abstract

This report presents the cleaning and analysis of a nutrition dataset collected by web scraping the publicly available food-composition website nutritionvalue.org, whose values originate from the USDA National Nutrient Database. The raw dataset contained 1,666 food records described by twelve attributes covering serving size and the major macro- and micronutrients. A structured cleaning pipeline was applied: text normalisation, numeric-type enforcement, removal of physically impossible values, per-column missing-value treatment, and removal of a constant (zero-information) column. Physically impossible outliers were removed, while valid statistical outliers such as oils and dried fruits were retained. The cleaned dataset contains 1,609 records (1,544 unique food names) across 105 categories and eleven attributes, with no missing values and no duplicate rows. Exploratory analysis, seven visualisations, and twelve data-driven insights were then produced. Key findings include a strong positive correlation between calories and total fat ($r = 0.82$), a clear relationship between sugar and carbohydrate content ($r = 0.77$), and large differences in energy density across categories, with Vegetable & Cooking Oils the most calorie-dense (~ 788 kcal/100 g) and Water the lowest (~ 1 kcal/100 g).

1. Introduction

Nutrition analytics applies data-science techniques to food-composition data in order to support dietary planning, public-health research, and medical decision-making. Reliable analysis depends on a clean and consistent dataset, and data collected from the web is rarely ready for immediate use: it often contains missing fields, duplicated entries, inconsistent formatting, and erroneous values introduced during collection. Food records were gathered from nutritionvalue.org — a public nutrition-facts website sourced from the USDA National Nutrient Database — into a raw dataset of 1,666 records and twelve columns: Food Name, Food Category, Serving Size (g), Calories (kcal), Protein (g), Total Fat (g), Carbohydrates (g), Fiber (g), Sugar (g), Sodium (mg), Cholesterol (mg), and Potassium (mg). The objectives are: (i) to inspect and understand the raw dataset, (ii) to design and apply an appropriate, well-justified data-cleaning pipeline, (iii) to verify that the cleaned dataset is analysis-ready, and (iv) to derive meaningful, quantified insights and visualisations describing the nutritional structure of the data.

2. Literature Review

Food-composition databases are a long-standing foundation of nutrition science. The USDA National Nutrient Database for Standard Reference, now delivered through USDA FoodData Central, is one of the most widely used public references and reports nutrient values on a standard per-100-gram basis to allow fair comparison between foods [1]. Consumer-facing platforms such as nutritionvalue.org re-present this public-domain data in an accessible, per-food format [2]. Because such data is published as web pages rather than analysis-ready tables, web scraping is a common data-collection strategy, and the BeautifulSoup library is a standard Python tool for parsing HTML and extracting structured fields [3]. Once collected, the data-cleaning stage — handling missing values, correcting types, and treating outliers — is widely recognised as the most time-consuming yet essential phase of any analytical project, and is well supported by the pandas library [4]. Exploratory data analysis and visualisation, now routinely performed with matplotlib [5] and seaborn [6], turn a cleaned dataset into interpretable insight. This project follows that established sequence: scrape, clean, explore, and interpret.

3. Related Work

Several categories of prior work relate to this project. Official food-composition resources such as USDA FoodData Central provide curated nutrient data but are intended as reference tables rather than teaching datasets, and still require reshaping and cleaning before category-level analysis [1]. Publicly shared nutrition datasets on data-science platforms are convenient but frequently mix reporting bases (per-serving versus per-100-gram) and carry their own duplication and missing-value problems; the brief for this work therefore restricted the source to a single website to keep the data consistent and traceable. General data-cleaning and exploratory-analysis workflows built on pandas, matplotlib and seaborn are commonly demonstrated on tabular datasets [4][5][6]. This project is distinct in applying such a workflow end-to-end to a self-collected, web-scraped nutrition dataset, and in emphasising domain-aware cleaning decisions — for example, treating a nutrient absent from a food label as zero rather than as a statistical gap, and distinguishing physically impossible values from legitimately extreme but valid foods.

4. Methodology

4.1 Data Source and Collection

Data was collected exclusively from nutritionvalue.org using the BeautifulSoup library. Listing pages were crawled to build a set of food detail-page links, and each detail page was parsed to extract the food name, category, serving size, and nutrient values. All values were normalised to a per-100-gram basis so that every record shares a single, comparable reporting unit, producing a raw dataset of 1,666 records and twelve columns.

4.2 Tools

The analysis was performed in Python within Google Colab, using pandas and NumPy for data manipulation and matplotlib and seaborn for visualisation.

4.3 Data-Cleaning Pipeline

The following steps were applied in order, each only where genuinely required:

- 1. Text cleaning:** leading/trailing and repeated spaces removed in Food Name and Food Category; meaningful capitalisation preserved.
- 2. Numeric-type enforcement:** all nutrient columns confirmed numeric; any non-numeric entry coerced to a missing value.
- 3. Impossible-value removal:** since values are per 100 g, any solid macronutrient exceeding 100 g per 100 g, or any food exceeding ~900 kcal per 100 g, is physically impossible and was removed as a collection error.
- 4. Missing-value treatment (per column):** rows missing a core nutrient (Calories, Protein, Total Fat, Carbohydrates) were dropped; Fiber, Sugar, Sodium and Cholesterol — nutrients absent from a label — were set to 0; Potassium, missing in ~68% of records, was estimated using each food category's median.
- 5. Unnecessary-column removal:** Serving Size (g) held the constant value 100 for every record and was removed.
- 6. Outlier detection:** outliers were detected with the IQR method but retained, as they represent valid foods (e.g. oils, dried fruit) rather than errors.

Exact duplicate rows were checked; none were present. Near-duplicate entries representing the same product in different pack sizes were deliberately kept as valid separate records.

5. Result Analysis

5.1 Dataset Overview and Structure

The cleaned dataset contains 1,609 rows and 11 columns — two text columns (Food Name, Food Category) and nine numeric (float64) columns, as confirmed by `info()`. Food Name has 1,544 unique values and Food Category has 105 unique values. The first and last records were inspected using `head()` and `tail()` to verify that the records were loaded correctly after cleaning.

5.2 Missing Values

After cleaning, every one of the eleven columns reported zero missing values (0.0%), confirming that the per-column missing-value treatment was fully successful.

5.3 Duplicates

The dataset contained 0 exact duplicate rows, so no duplicate removal was required. Near-duplicate products (the same item in a different pack size) were intentionally kept as distinct records.

5.4 Cleaning Outcome

The pipeline reduced the dataset from 1,666 records and twelve columns to 1,609 records and eleven columns, removing 57 rows (21 physically impossible values and 36 rows missing a core nutrient) and one constant column.

Stage	Value
Raw records	1,666
Impossible rows removed	21
Core-missing rows removed	36
Constant column removed	1 (Serving Size)
Final records	1,609
Final columns	11
Missing values (final)	0
Duplicate rows (final)	0

5.5 Descriptive Statistics — describe()

The describe() output gives count, mean, standard deviation, minimum, quartiles, and maximum for every numeric column (the 50% column is the median), so it also covers the mean/median/min/max/standard-deviation summary.

Nutrient	count	mean	std	min	25%	50%	75%	max
Calories (kcal)	1609	247.08	189.14	0	60	218.8	392.9	900
Protein (g)	1609	15.73	17	0	2.35	9.58	23.81	86.96
Total Fat (g)	1609	12.62	15.99	0	0	6.25	21.43	100
Carbohydrates (g)	1609	27.41	27.34	0	4.22	14.29	51.11	100
Fiber (g)	1609	2.72	4.9	0	0	0.9	3.51	40
Sugar (g)	1609	14.43	20.28	0	0	3.87	22.83	100
Sodium (mg)	1609	30.25	264.71	0	1.19	10	22.54	7246.67
Cholesterol (mg)	1609	6.55	11.21	0	0	0	10	71.43
Potassium (mg)	1609	5.91	8.48	0	2.04	4.85	7.14	206.45

5.6 Category Frequency — value_counts()

Across 105 categories, Cheese is the most common (159 foods) and several categories (e.g. Gravy Mix) contain only one. The ten most frequent categories are:

Food Category	Number of Foods
Cheese	159
Pre-Packaged Fruit & Vegetables	115
Pickles, Olives, Peppers & Relishes	92
Cookies & Biscuits	84

Food Category	Number of Foods
Chocolate	67
Popcorn, Peanuts, Seeds & Related Snacks	55
Dips & Salsa	44
Other Drinks	44
Candy	39
Cooked & Prepared	37

5.7 Group-Based Analysis — groupby()

Grouping by category, considering only categories with at least ten foods for reliability, reveals substantial differences in average calories per 100 g. The ten most calorie-dense categories are shown below, together with their average protein and sugar values. By average protein, Cheese leads (~46.6 g/100 g), followed by unprocessed meat/poultry (~40.7 g/100 g) and gelatin-based desserts (~37.5 g/100 g).

Food Category	Foods	Avg Cal	Avg Protein	Avg Sugar
Vegetable & Cooking Oils	11	788	2	0.2
Chocolate	67	525.2	13.2	38.3
Chips, Pretzels & Snacks	31	513.5	14.2	6.6
Baking Decorations & Dessert Toppings	12	492.9	16.8	17.2
Popcorn, Peanuts, Seeds & Related Snacks	55	471.4	22.2	30.1
Cookies & Biscuits	84	461.4	10.8	37.9
Other Snacks	20	454.7	35.2	19.1
Snack, Energy & Granola Bars	26	438.2	35.4	26.7
Cereal	24	400.3	24.7	18.8

Food Category	Foods	Avg Cal	Avg Protein	Avg Sugar
Cakes, Cupcakes, Snack Cakes	13	398.6	6.6	40.1

5.8 Correlation Matrix — corr()

The correlation matrix of the nine numeric columns is shown below (Cal = Calories, Prot = Protein, Fat = Total Fat, Carb = Carbohydrates, Sod = Sodium, Chol = Cholesterol, Pot = Potassium).

	Cal	Prot	Fat	Carb	Fiber	Sugar	Sod	Chol	Pot
Cal	1	0.42	0.82	0.56	0.28	0.41	-0.01	0.29	0.19
Prot	0.42	1	0.38	-0.06	0.21	-0.11	0.07	0.61	0.09
Fat	0.82	0.38	1	0.04	0.18	0	-0.02	0.38	0.08
Carb	0.56	-0.06	0.04	1	0.26	0.77	0	-0.23	0.24
Fiber	0.28	0.21	0.18	0.26	1	-0.02	0.01	-0.27	0.53
Sugar	0.41	-0.11	0	0.77	-0.02	1	0.01	-0.15	0.07
Sod	-0.01	0.07	-0.02	0	0.01	0.01	1	-0.02	0.01
Chol	0.29	0.61	0.38	-0.23	-0.27	-0.15	-0.02	1	-0.21
Pot	0.19	0.09	0.08	0.24	0.53	0.07	0.01	-0.21	1

Applying a threshold of $|r| \geq 0.6$, the strong relationships are: Calories & Total Fat ($r = 0.82$), Carbohydrates & Sugar ($r = 0.77$), and Protein & Cholesterol ($r = 0.61$). All three are positive; no strong negative relationship was found.

5.9 Outlier Detection — IQR

The IQR method reports the following outlier counts per column. These were detected but retained, because they correspond to valid foods (e.g. oils and dried fruit) rather than errors; only the physically impossible values identified earlier were removed.

Nutrient	IQR Outliers (kept)
Calories (kcal)	1
Protein (g)	24
Total Fat (g)	25
Carbohydrates (g)	0
Fiber (g)	125
Sugar (g)	78
Sodium (mg)	56
Cholesterol (mg)	157
Potassium (mg)	75

5.10 Visualizations

Five charts summarize the key findings from the cleaned dataset.

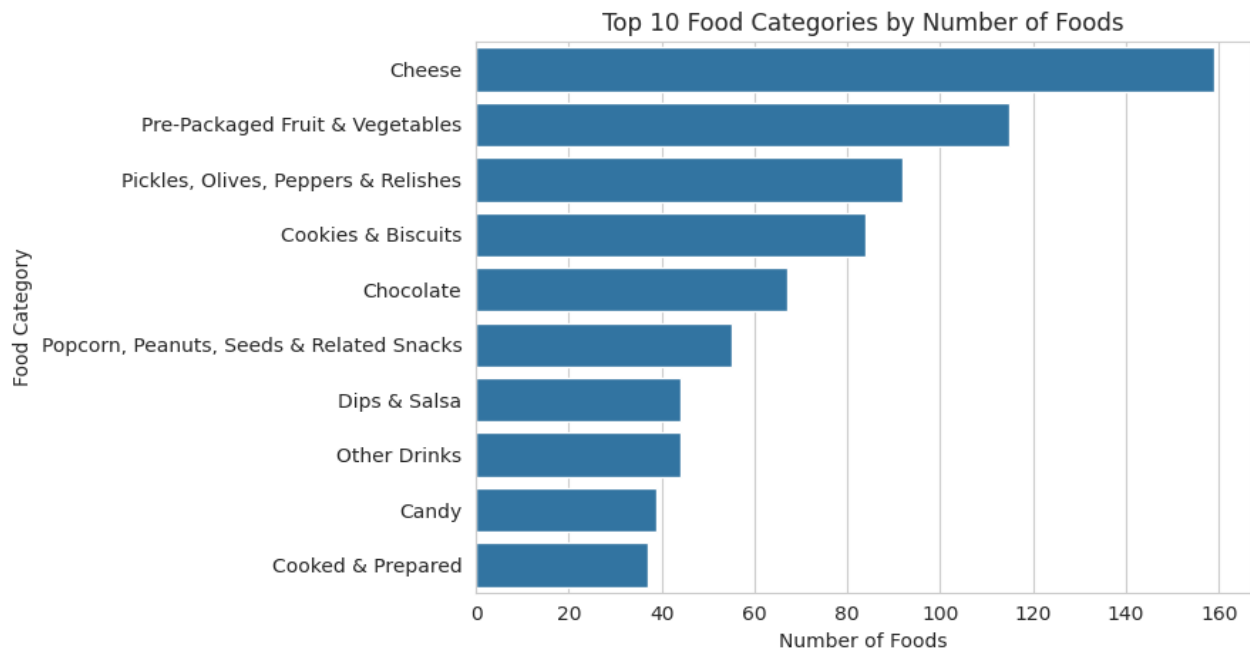


Figure 1 — Top 10 food categories by number of records.

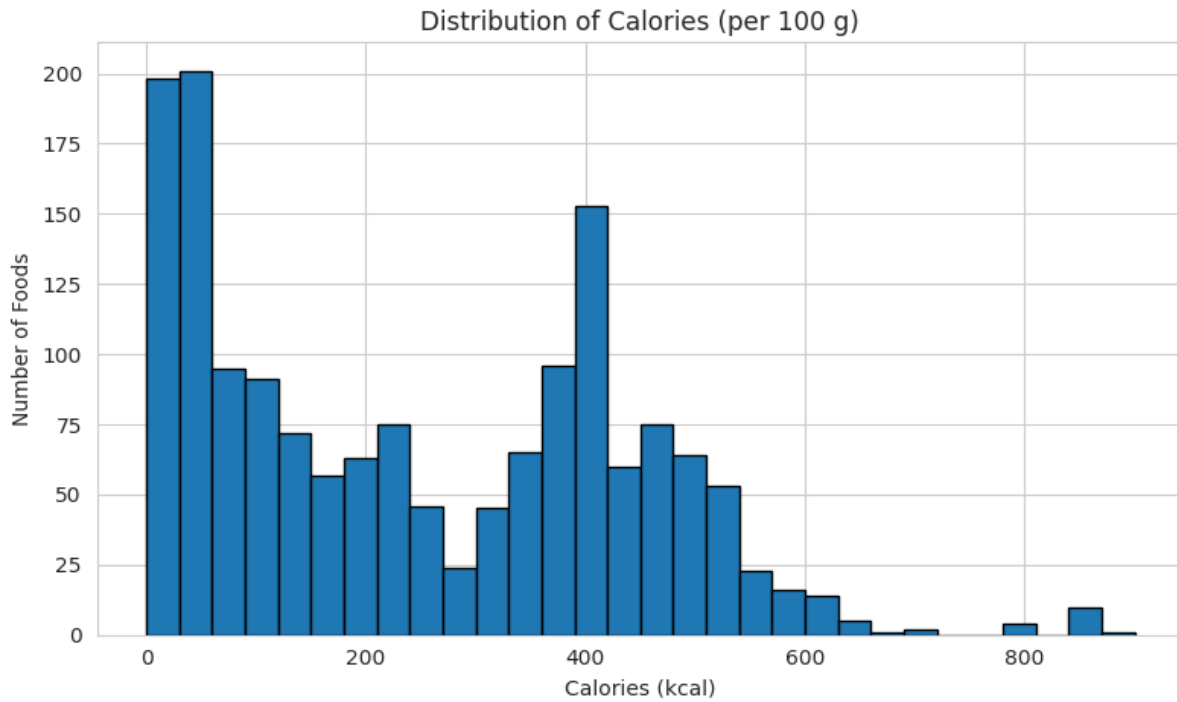


Figure 2 — Distribution of calories per 100 g (right-skewed).

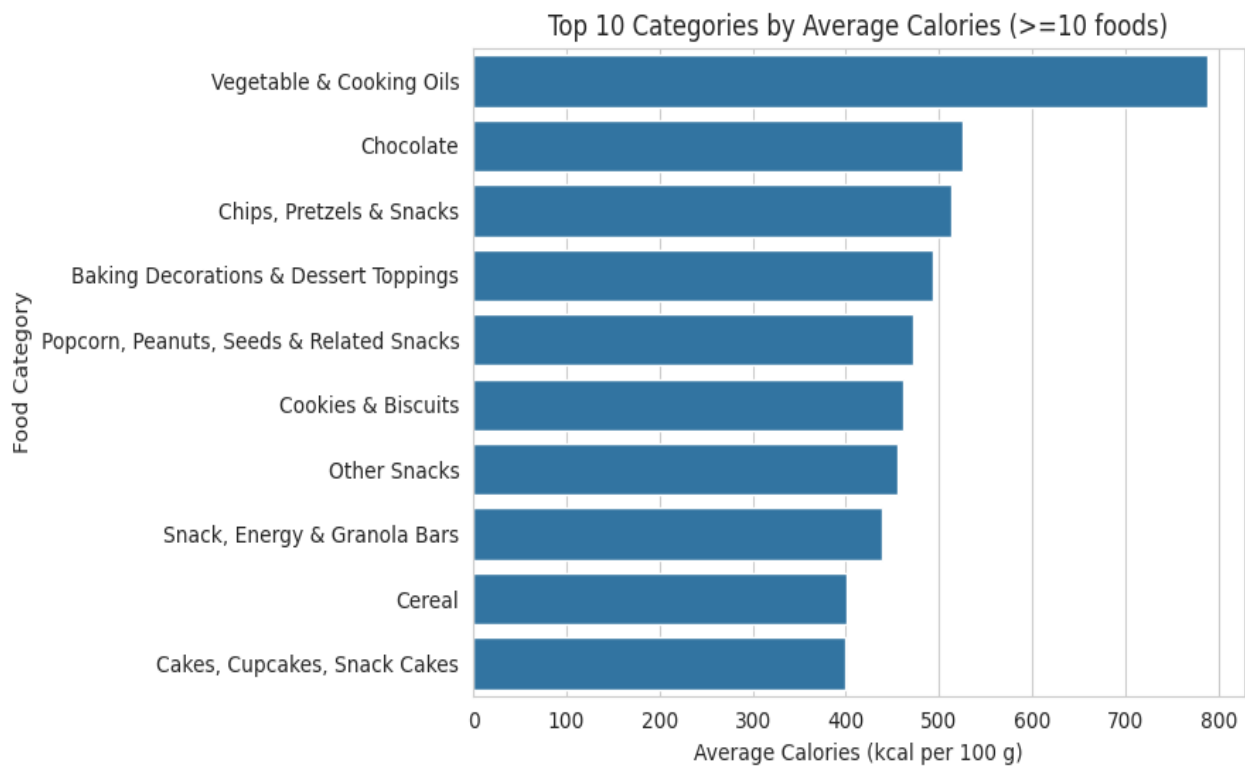


Figure 3 — Top 10 categories by average calories (≥ 10 foods).

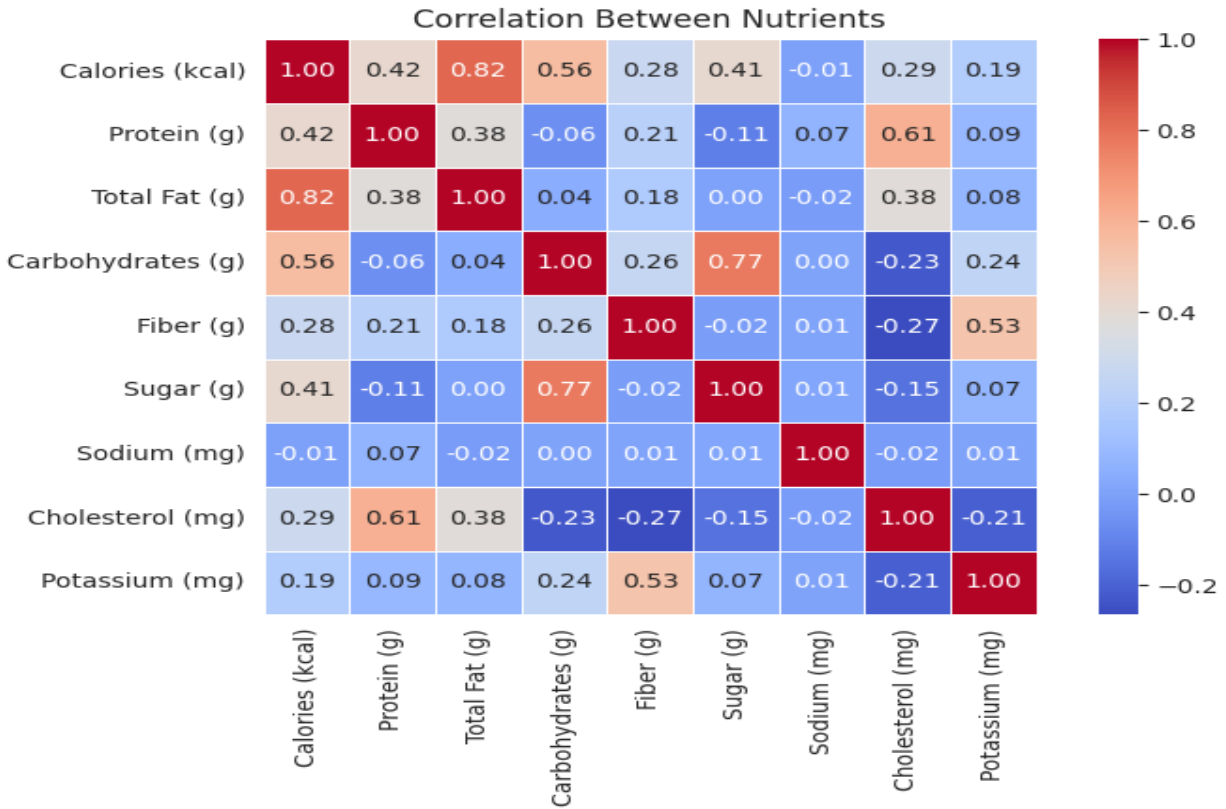


Figure 4 — Correlation heatmap of the nutrient columns.



Figure 5 — Calories vs Total Fat (strongest relationship, $r = 0.82$).

5.11 Key Insights

The following data-driven insights were calculated directly from the cleaned dataset:

1. The cleaned dataset contains 1,609 foods (1,544 unique names) across 105 categories, described by 11 nutritional attributes.
2. Cleaning removed 57 rows in total and one constant column, leaving 0 missing values and 0 duplicate rows.
3. Cheese is the most common category (159 foods); the rarest (e.g. Gravy Mix) contains only one.
4. The average energy content is 247.1 kcal per 100 g, with a median of 218.8 kcal per 100 g (a right-skewed distribution).
5. Vegetable & Cooking Oils is the most calorie-dense category (~788 kcal/100 g); Water is the lowest (~1 kcal/100 g).
6. The single most calorie-dense food reaches 900 kcal per 100 g (a cacao-butter product).
7. Cheese is the highest-protein category on average (~46.6 g/100 g); the highest-protein single food reaches 87.0 g per 100 g.
8. Gelatin, Gels, Pectins & Desserts is the sweetest category on average (~81.2 g sugar/100 g); the highest-sugar food reaches 100 g per 100 g.
9. Calories correlate most strongly with total fat ($r = 0.82$), more than with carbohydrates ($r = 0.56$) or protein ($r = 0.42$).
10. Sugar and carbohydrates are strongly correlated ($r = 0.77$), consistent with sugar being a type of carbohydrate.
11. Protein and cholesterol are moderately correlated ($r = 0.61$), reflecting animal-protein foods such as cheese and meat.
12. About two-thirds of Potassium values were estimated during cleaning, so Potassium-based conclusions should be treated with caution.

6. Conclusion

This project delivered a complete nutrition-analytics workflow, transforming a raw web-scraped dataset of 1,666 records into a clean, consistent, analysis-ready dataset of 1,609 records with eleven attributes, no missing values, and no duplicates. Domain-aware cleaning decisions — distinguishing impossible values from valid outliers and choosing per-column missing-value strategies — preserved the integrity of the data while making it usable. The analysis produced quantified, data-driven insights: fat is the dominant driver of energy density, sugar and carbohydrates are closely linked, and energy density varies enormously across categories from oils to water. The main limitation is the Potassium column, which was largely estimated and should be interpreted cautiously. Future work could expand the dataset across all alphabetical and category listings to increase coverage, and could extend the analysis to dietary-recommendation or food-classification models built on the cleaned data.

References

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